

HTML (HyperText Markup Language)

What is HTML?

HTML (HyperText Markup Language) was created by Tim Berners-Lee in 1991 as a standard for creating web pages. It's a markup language used to structure content on the web, defining elements like headings, paragraphs, links, and images. HTML forms the backbone of web content. In layman's terms, HTML is like the skeleton of a website. It's a set of instructions that tells a web browser how to display text, images, videos, and other elements on a webpage. Think of it as the building blocks that create the structure and look of a website, similar to how bricks and mortar are used to build a house.

In a nutshell:

- HTML is the language of the web, used to create websites.
- HTML defines the barebone structure or layout of web pages that we see on the Internet.
- HTML consists of a set of tags contained within an HTML document, and the associated files typically have either a ".html" or ".htm" extension.
- There are several versions of HTML, with HTML5 being the most recent version.

Features of HTML

- It is platform-independent. For example, Chrome displays the same pages identically across different operating systems such as Mac, Linux, and Windows.
- Images, videos, and audio can be added to a web page (For example - YouTube shows videos on their website)
- HTML is a markup language and not a programming language.
- It can be integrated with other languages like CSS, JavaScript, etc. to show interactive (or dynamic) web pages

Why the Term HyperText & Markup Language?

The term 'Hypertext Markup Language' is composed of two main words: 'hypertext' and 'markup language.' 'Hypertext' refers to the linking of text with other documents, while 'markup language' denotes a language that utilizes a specific set of tags.

Thus, HTML is the practice of displaying text, graphics, audio, video etc. in a certain way using special tags.

Note: Tags are meaningful texts enclosed in angle braces, like '<...>'. For example, the '<head>' tag. Each tag has a unique meaning and significance in building an HTML page, and it can influence the web page in various ways.

A beautiful analogy to understand HTML, CSS, and JavaScript:



Car Skeleton (only
body) is **HTML**

Car Painted or
Decorated is **CSS**

Car Engine and
Internal logic is **JS**

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In building a webpage, think of HTML, CSS, and JavaScript as different parts of a car. HTML is like the car's skeleton, forming the basic structure and frame. CSS adds the paint and finishing touches, making the car look appealing with color, style, and design. JavaScript is similar to the engine and mechanical parts, infusing the car with functionality, movement, and interactive features. Similarly, when developing a website, HTML lays out the structure, CSS enhances its visual appeal, and JavaScript provides interactivity and dynamic content

History of HTML:

- In 1989, Tim Berners-Lee established the World Wide Web (www), and in 1991, he created the first version of HTML.
- From 1995 to 1997, further work was done to develop and refine different versions of HTML.
- In 1999, a committee was organized that standardized HTML 4.0, a version still used by many today.
- The latest and most stable version of HTML is 5, also known as HTML5.

Feel free to read more history of HTML here on the [HTML Wikipedia page](#) but I will move ahead and cover important aspects of HTML.

HTML Working

You must have heard of frontend and backend. Frontend refers to the visible part of a website or app that users interact with, like the tables, images, and buttons. It's built using languages like HTML, CSS, and JavaScript. The backend, on the other hand, handles behind-the-scenes operations like storing and processing data when users interact with the frontend. It uses languages like Python, Ruby, and Java. In essence, frontend is what users see, while backend manages the functionality.

How do websites work?

When we want to access any information on the internet, we search for it using a web browser. The web browser retrieves the content from web servers, where it is stored in the form of HTML documents.

An HTML document is created by writing code with specific tags in a code editor of your choice. The document is then saved with the **'html'** extension. Once saved, the browser interprets the HTML document, reads it, and renders the web page.



The text editor has the HTML code of a website. This website can now be viewed in a beautifully rendered format using a computer program known as a web browser.

What is a Web Browser?

A web browser is a program that understands HTML tags and renders them in a human-readable format that is easily viewable by people visiting the website. Developers write code in HTML because it's a straightforward way to instruct the web browser on what to display. In the next section, I'll show you how to set up VS Code for writing your own HTML code and rendering it in a web browser.

What is an HTML Document?

An HTML document is a text document saved with the **'html'** or **'html'** extension, containing text and specific tags enclosed in **'< >'**. These tags provide the necessary instructions for configuring the web page. The tags themselves are standardized and fixed. The structure of an HTML document will be explained later in this HTML tutorial.

What is a Rendered Page:

The rendered page is the output screen of our HTML Document which is the page displayed on the browser.

How does a basic website work?

- Web Browser(client) requests websites like www.facebook.com from the web server.
- Web server in return sends HTML, CSS, and JavaScript files to it.
- HTML, CSS, and JavaScript files are parsed by a web browser which is responsible for showing you a beautiful website.

How does a browser work?

A web browser plays a crucial role in parsing and rendering HTML code to the client. A web browser is a complex program that performs many tasks behind the scenes. Here's a summary of how it works:

- A browser is an application that reads HTML documents and displays them as web pages. Browsers can't access the content directly from where it's stored; this is where servers come into play.
- A server acts as an intermediary, listening to browser requests and fulfilling them by delivering the HTML document to the browser.
- Web browsers perform two main tasks: parsing and rendering.
- During the parsing stage, the browser receives raw bytes, which are converted into characters. These characters are then converted into tokens, which in turn are transformed into nodes. These nodes are organized into a tree-like data structure known as the DOM (Document Object Model).
- Once the DOM tree is constructed, the browser moves on to the rendering stage. At this point, each node in the DOM tree is rendered and displayed on the screen.

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Installation & Setup of Visual Studio Code for HTML:

We will install and set up HTML to optimize its utility for creating web pages. Additionally, we'll install extensions in Visual Studio Code to enhance its functionality. If you're unsure about which editor to use, you can confidently start with Visual Studio Code. You won't regret it; it's one of the best free code editors available in the market.

- Search for "Visual Studio Code download" on Google
- Download [Visual Studio Code](#) for your Operating System. I am using Windows so I will install it for Windows

Here is a quick video showing VS Code Installation:

Live Server Extension

In addition, we'll be installing the Live Server extension in our Visual Studio Code editor to view live reload pages.

The Live Server extension is used to launch a local development server with a live reload feature for HTML pages.

Your Journey to Creating Your First Website Begins Here!

Let's mark this as an important milestone: the creation of your first website! And what's a better way to start than with the traditional "Hello, World!"?

Creating a New File

Click on "Open Folder" and open a folder somewhere on your computer. I am opening a folder named html-tutorial

Once VS Code is open, you'll want to create a new file:

1. Click on the "New File" icon in VS Code.
2. Type the filename as "index.html" and hit Enter.

Now that your file is ready, copy the following code and paste it into your "index.html" file.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Document</title>
```

```
</head>
```

```
<body>
```

```
  Hello World
```

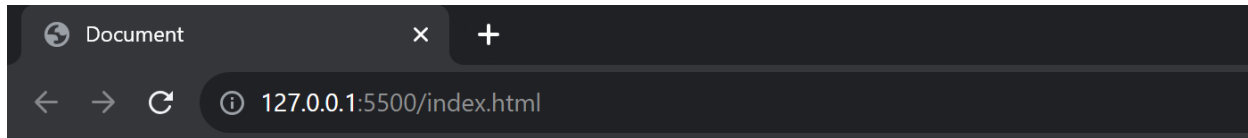
```
</body>
```

```
</html>
```

To see your webpage in action, locate the "Go Live" icon at the bottom-right corner of your VS Code window and click it. If you don't see this icon, you probably haven't installed the Live Server extension.

Your First Website is Live!

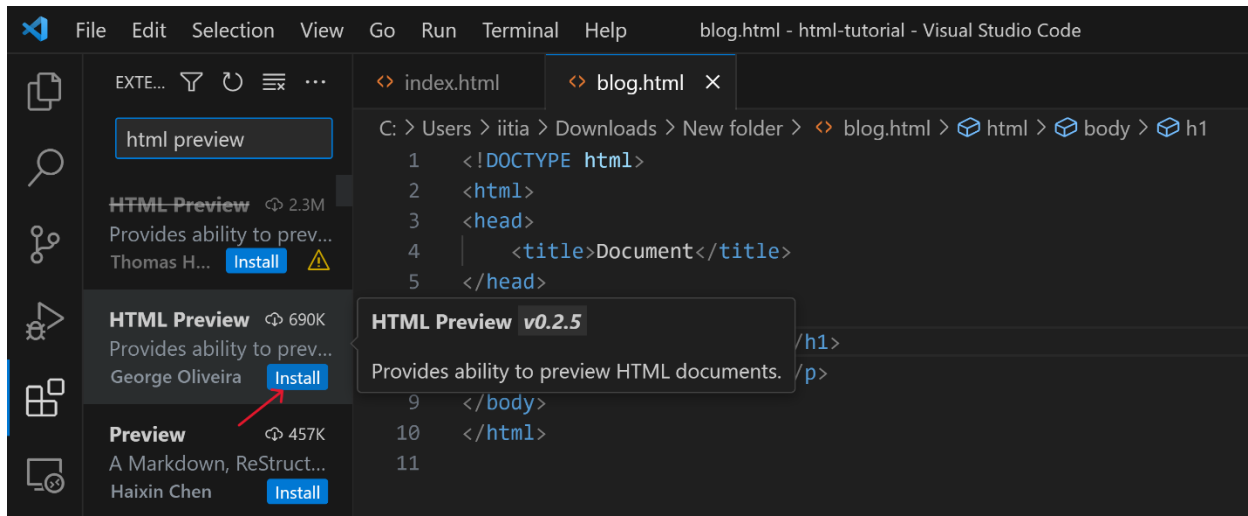
Congratulations! If you've followed along, you should now see your very first website displaying the classic "Hello, World!" message.



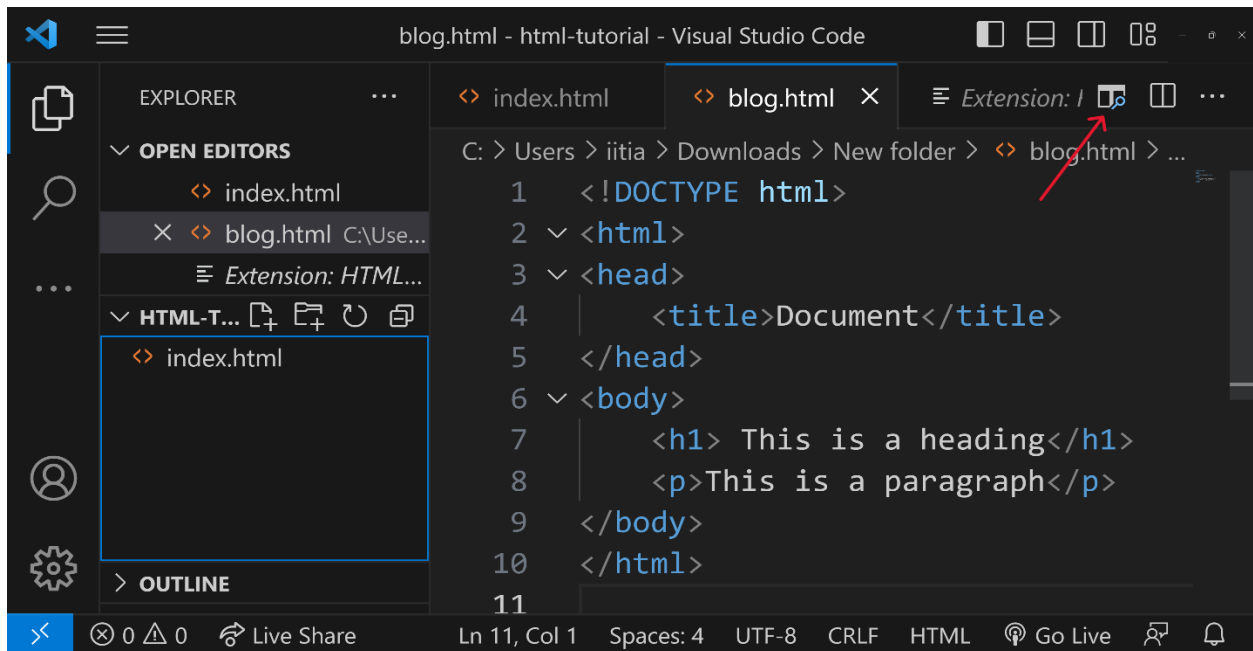
Hello World

Live Preview Extension

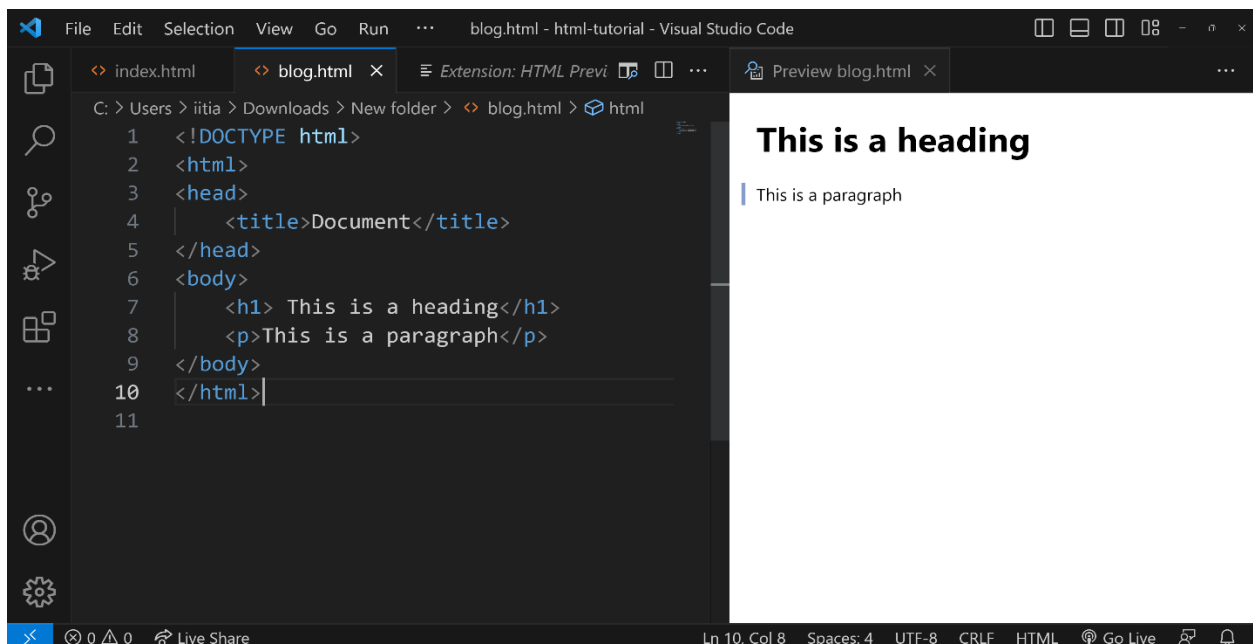
Another useful extension for working with HTML in VS Code is 'HTML Preview.' To install it, simply click on the extensions icon in VS Code and type 'HTML Preview' in the search bar. Install it!



Now, you will see a button within VS Code. Clicking on this button will allow you to preview your HTML right within the editor.



Once you click the button, you'll see a live preview of your HTML directly within VS Code.



You don't even need a browser to render plain HTML. This live preview feature in VS Code is perfect for this HTML tutorial, allowing you to build and preview your entire HTML website without ever leaving the editor.

HTML Page Structure

An HTML document is structured using a set of nested tags. Each tag is enclosed within `<...>` angle brackets and acts as a container for content or other HTML tags. Let's take a look at a basic HTML document structure:

```
<!DOCTYPE html>

<html>

<head>

  <title>Document</title>

</head>

<body>

  <!-- content -->

</body>

</html>
```

A typical HTML page looks like this:

```
<html>

  <head>

    <title>Page title</title>

  </head>

  <body>

    <h1>This is a heading</h1>

    <p>This is a paragraph.</p>

    <p>This is another paragraph.</p>

  </body>

</html>
```

Almost every website uses this structure. The main content goes inside the body tag. No worries if this looks complicated; let's break it down!

Note: These are the essential elements for a basic HTML document: **<!DOCTYPE html>**, **<html>**, **<head>**, **<title>**, **</head>**, **<body>**, **</body>**, **</html>**

DOCTYPE Declaration

```
<!DOCTYPE html>
```

The `<!DOCTYPE html>` declaration informs the web browser about the HTML version being used. The latest version is HTML5. But if this changes in the future (maybe 10 years down the line), the doctype declaration will be helpful!

HTML Root Element

```
<html>
```

The `<html>` tag is the root element that encapsulates all the content on the page.

```
</html>
```

The `</html>` tag marks the end of the `<html>` section.

Head Section

```
<head>
```

The `<head>` tag contains metadata and links to external resources like CSS and JavaScript files.

```
</head>
```

The `</head>` tag marks the end of the `<head>` section.

Title Tag

```
<title>Document</title>
```

The `<title>` tag sets the title of the web page, which is displayed in the browser's title bar or tab.

Body Tag

```
<body>
```

The `<body>` tag contains the visible content of the web page. This is where text, images, and other elements go.

```
</body>
```

The `</body>` tag marks the end of the visible content of the web page.

Every HTML page should include at least these essential elements to define the basic layout. In upcoming tutorials, we'll dive deeper into the fascinating world of HTML.

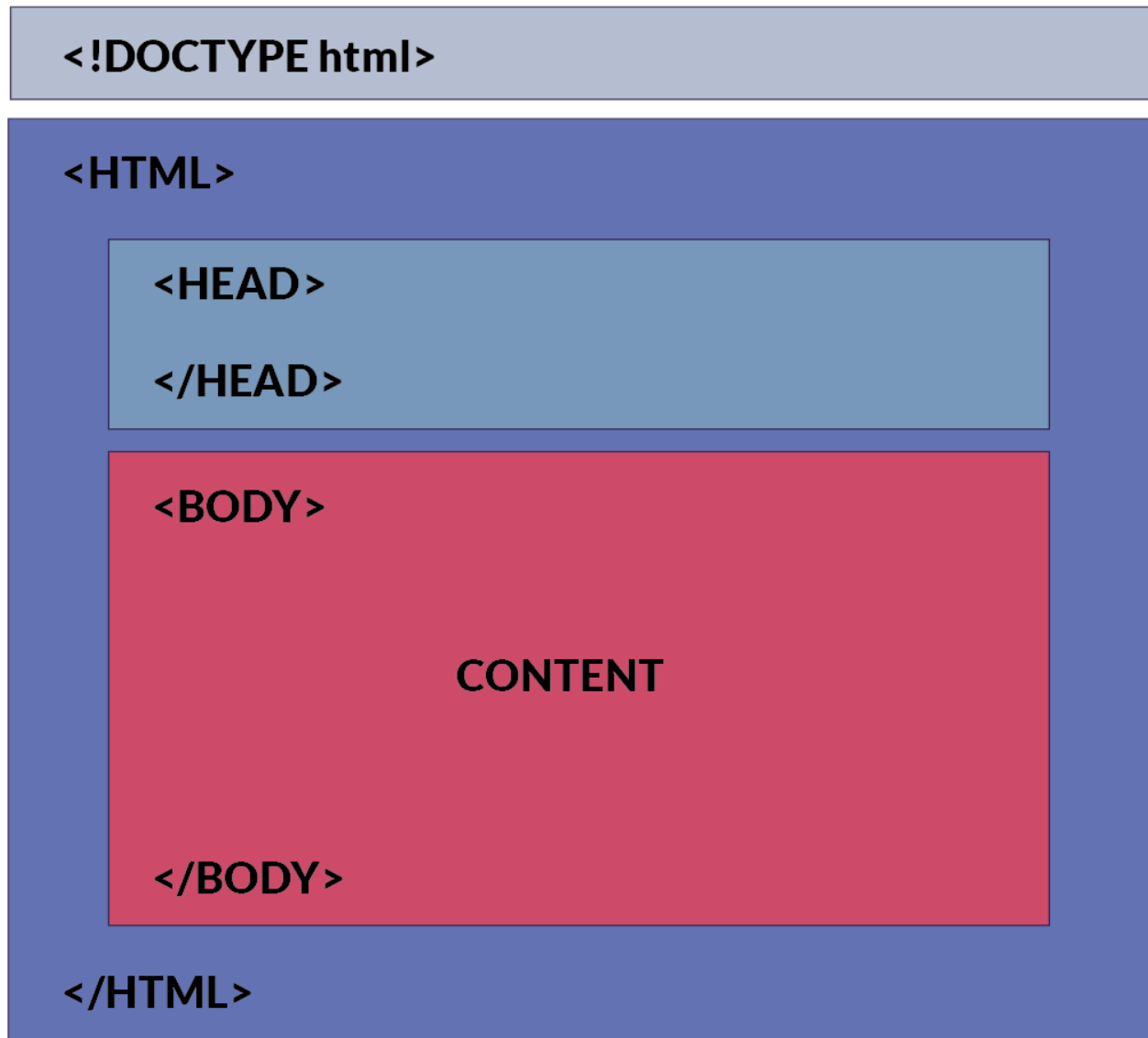
Summary

- The `<!DOCTYPE html>` tag specifies that the document is an HTML5 document.
- The `<html lang="en">` tag defines the document to be in English.

- The <head> section contains metadata and the title of the webpage, which appears in the browser's title bar.
- The <body> section contains the content that will be displayed on the webpage.
- The h1 and p are two types of tags. We will learn about more tags in the later section

Visualization of an HTML Document:

The following image provides a visual representation of the HTML structure:



How This Content Appears in a Web Browser:

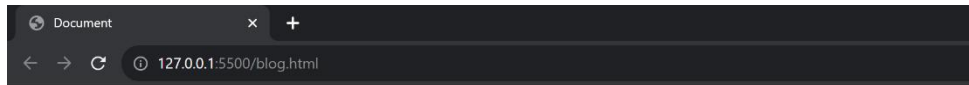
Consider this html code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
  <title>Document</title>
</head>
<body>
  <h1> This is a heading</h1>
  <p>This is a paragraph</p>
</body>
</html>
```

Below is an image showing how this HTML document will be rendered in a web browser:



This is a heading

This is a paragraph

In the browser, the title bar will display the content from the `<head>` section, specifically the `<title>` tag. The main area of the browser window (usually a white background) will display the content inside the `<body>` tag.